



CyberLLaMA: A fine-tuned large language model for cybersecurity named entity recognition

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ABSTRACT

Cybersecurity-specific Named-Entity Recognition (NER) is critical in addressing the escalating complexity and evolving cyber threats. While deep learning methods form the foundation of modern NER tasks, they still fall short in addressing cybersecurity-specific NER tasks due to the limited availability of up-to-date cybersecurity datasets and the unique characteristics of cybersecurity terminology (jargon, abbreviations, and rapidly evolving vocabulary). To bridge this gap we (i) compile a 42 404-article corpus and manually annotate 4788 unique security terms, and (ii) present **CyberLLaMA**-a framework that fine-tunes LLAMA-3.2-3B and stacks a bidirectional LSTM plus conditional-random-field layers to preserve label consistency. On the held-out test set, CyberLLaMA attains an F_1 of 98.88 %, surpassing RoBERTa, SCBERT, and GPT-NER. The results indicate that **CyberLLaMA**, as an effective solution for Cybersecurity NER tasks, offers practical value to cybersecurity professionals and the general public by enhancing the extraction of cybersecurity information in texts.

1. Introduction

Driven by increasingly complex network environments and recent advances in cloud computing [1], the formidable challenge of keeping pace with rapid technological developments amid a flood of online data [2], and the rise of escalating cyber threats [3,4], cybersecurity has become a pressing global issue [5]. While innovative countermeasures-such as securing digital transactions [6] and query authentication schemes aimed at protecting inference privacy [1]-are actively being developed, enhancing public cybersecurity awareness and correcting persistent misconceptions remains imperative [7]. Meanwhile, as online articles from reliable sources serve as a significant channel for the general public to access information, they play a crucial role in achieving this [8]. However, issues like jargon, abbreviations, and rapidly evolving vocabulary, along with the overlap of many cybersecurity terms with everyday language [9] may confuse the readers in understanding their meanings. This issue is further aggravated by the global cybersecurity workforce shortage [10]. In this context, Named Entity Recognition (NER) has garnered increasing attention as a crucial tool for automatically identifying and classifying cybersecurity-specific entities in textual data.

Previous research has introduced various models and architectures to address the performance of domain-specific NER tasks in different domains. Du et al. proposed an MRC-based nested medical NER model incorporating adaptive pretraining to improve medical-specific NER tasks [11]. Zhong et al. developed a BERT-Span model trained with Chinese rehabilitation medicine corpus to extract information in the domain of rehabilitation medicine [12]. Zhang et al. proposed FinBERT-MRC, which adapted the MRC framework with BERT for the domain of financial NER to address the challenges of recognizing financial entities in texts [13]. Additionally, Tang et al. presented DoSEA, a domain-specific entity-aware framework for robust cross-domain NER to solve problems caused by certain entity types from specific domains [14]. However, the recognition of cybersecurity entities from literature still faces notable challenges. First, security-related literature often contains domain-specific jargon and abbreviations. Second, the rapid evolution of the cybersecurity lexicon, including new terms and attack techniques [15], could make the cybersecurity training datasets outdated. Meanwhile, entities in cybersecurity literature can carry different meanings in general contexts. In addition, security-related literature frequently involves complex contextual dependencies [9], which further increases the difficulties.

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With the recent breakthrough of the large language models (LLMs) like ChatGPT [16], LLaMA [17], Claude [18], and others, the field of natural language processing (NLP) has achieved unprecedented advancements. These models, equipped with a large number of parameters and trained on extensive datasets, achieved outstanding achievements across a range of NLP tasks but surprisingly underperformed on Named Entity Recognition (NER) tasks when compared to previous benchmarks [19]. Using large language models (LLMs) for Named Entity Recognition (NER) also presents several challenges. First, LLMs are primarily pre-trained on large amounts of corpora across all domains and tend to under-perform in domain-specific NER tasks [20]. This limitation is particularly notable in cybersecurity tasks due to its technical lexicon and rapidly evolving vocabulary [21]. Second, as sentences in cybersecurity texts are frequently long and complex [22], their inherent structures also demand more sophisticated approaches to effectively capture the nuanced patterns in cybersecurity terminology [23]. Moreover, in any NER system, token labels must form a sequence that adheres to specific structural constraints [24]. In our case, this is demonstrated by the BIO annotation scheme, which dictates that an “I-security” tag should not appear before a “B-security” tag (see Section 3.3 for details). When LLMs are directly applied to NER tasks, they can generate inconsistent label sequences, resulting in fragmented entities [19].

Previous research indicates that high-quality annotated corpora are essential for NER tasks, as they provide the foundational training material and reliable evaluation benchmarks that allow models to learn complex patterns [22]. Especially for cybersecurity-specific NER tasks, an up-to-date cybersecurity-specific dataset is crucial for overcoming the unique challenges posed by cybersecurity texts like rapidly evolving terminology and highly technical descriptions [25]. However, there remains a notable gap, that a dataset meets the previous requirement is still missing, hindering progress in developing up-to-date cybersecurity-specific NER systems.

In addressing all these aforementioned challenges, we first propose **CyberLLaMA**, a novel framework that integrates a Large Language Model (LLM) fine-tuned from LLaMA, with the addition of the Bidirectional LSTM (BiLSTM) layer [26–29] and the Conditional Random Field (CRF) layer [30]. The fine-tuning process is employed to tailor the model to meet domain-specific requirements. Concurrently, the incorporation of the BiLSTM layer enables the model to capture the long-range dependencies intrinsic to complex text sequences, while the integration of the CRF layer further refines the modeling of sequential dependencies, thereby substantially enhancing entity recognition performance. By preserving LLaMA’s inherent efficiency and versatility and incorporating these targeted enhancements, CyberLLaMA effectively addresses the unique challenges of cybersecurity-specific Named Entity Recognition (NER).

Meanwhile, we introduce a dataset that comprises 42,404 up-to-date articles that are related to cybersecurity from a diverse set of sound sources. From these articles, we extracted and manually annotated 4788 unique cybersecurity terms using a rigorous annotation process that combined automated extraction techniques with expert review. Our experimental evaluation demonstrates that CyberLLaMA achieves superior performance compared to other models. After fine-tuning on a comprehensive security-specific dataset, CyberLLaMA achieved an F1 score of 98.88%, a precision of 98.62%, and a recall of 99.13%, significantly outperforming other baseline models. These results highlight the robustness of our approach and its effectiveness in accurately identifying and classifying security-related entities, demonstrating its potential as a powerful tool for cybersecurity-specific Named Entity Recognition (NER) tasks.

The main contributions and novelties of our work are summarized as follows:

- We established an updated cybersecurity-specific dataset comprising 42,404 cybersecurity-related articles from newspapers, security blogs, and official websites, then based on that, collected and an-

notated 4788 unique cybersecurity terms. Together, they provide a robust foundation for cybersecurity-related Named Entity Recognition (NER).

- We propose **CyberLLaMA**, a novel framework that integrates a fine-tuned Large Language Model (LLM) with a Bidirectional LSTM (BiLSTM) and a Conditional Random Field (CRF) layer specifically designed for cybersecurity-specific Named Entity Recognition (NER). Experimental results demonstrate that CyberLLaMA significantly outperforms existing models, achieving state-of-the-art performance on cybersecurity-specific NER tasks.

2. Background

In this section, we briefly introduce Named Entity Recognition (NER) and then provide an overview of the NER task formulation and notation methods. We last explore the recent advancements in large language models (LLMs).

2.1. Named entity recognition (NER)

A named entity is a proper noun, whether a single word or a phrase, that serves as a name for something or someone [31,32]. Named Entities (NEs) can broadly be divided into two categories, generic named entities and domain-specific named entities [32]. For example, generic named entities include organization, person, and location names [32], while the domain-specific named entity examples include threats, vulnerabilities, malware, and attacks.

Named Entity Recognition (NER) is a fundamental task that focuses on identifying and categorizing entities within text into predefined categories that have similar attributes [31,33]. NER is a vital preprocessing step for various applications such as information retrieval, question answering, and machine translation [32].

Traditionally, NER has been framed as a sequence labeling problem, where each word or token is tagged with its corresponding entity label [34]. Early attempts at NER leveraged rule-based and statistical methods, but these approaches were often limited in their ability to generalize across different domains.

The introduction of deep learning marked a significant shift in the field, and has achieved success in multiple domains [35–39]. Recent advances have substantially improved NER tasks through techniques such as Conditional Random Fields (CRF) [30] and Bidirectional Long Short-Term Memory (BiLSTM) networks [26–29]. CRF is employed to predict entity labels by modeling sequential dependencies, ensuring that the label sequence is globally optimal [30]. Meanwhile, BiLSTM captures contextual information in both forward and backward directions [26,27], thereby enriching token representations for more accurate entity recognition. Hammerton [40] pioneered the use of unidirectional LSTMs for token-level representation, followed by Collobert et al. [41], who combined CNNs with Conditional Random Fields (CRF) for sequence tagging. Lample et al. [42] improved upon this by integrating bidirectional LSTMs with CRFs, a structure that became a cornerstone for many subsequent NER systems.

Character-level representations also gained prominence, with Chiu and Nichols [43] introducing character-level CNNs to capture morphological features, while Ma and Hovy [44] combined these with word-level embeddings to enhance predictions. The development of pre-trained language models further revolutionized NER, BERT [45] demonstrated significant improvements by leveraging bidirectional contextual embeddings, and its domain-specific variants, such as SciBERT [37] and BioBERT [38], showcased the utility of pretraining on specialized datasets.

Recent years have also witnessed a growing interest in reformulating NER as a machine reading comprehension (MRC) task. Li et al. [46] demonstrated that treating NER as an MRC task can address challenges like nested entities and overlapping spans by extracting entities as answer spans to specific queries. This paradigm shift highlighted the

adaptability of NER frameworks to complex tasks requiring deeper contextual understanding.

2.2. NER task formulation and annotation

High-quality annotations play a crucial role in achieving this [32]. Technically, given a sequence of input tokens (words or subwords):

$$X = (x_1, x_2, \dots, x_n),$$

and the goal of NER is to identify contiguous subsequences corresponding to named entities, and assign each subsequence a categorical entity label (e.g., PERSON, ORGANIZATION, LOCATION). Historically, NER has often been formulated as a *sequence labeling problem*. Under this formulation, the model generates a sequence of labels:

$$Y = (y_1, y_2, \dots, y_n),$$

where each y_i corresponds to the token x_i [47] and drawn from a predefined set of tags. A popular labeling scheme is BIO tagging, in the BIO scheme:

- B-type: indicates the beginning of an entity of a given type (e.g., “B-security” for the first token of a security-related entity).
- I-type: indicates continuation (inside) of an entity (“I-security” for subsequent tokens of the same security-related entity).
- O: indicates a token outside any entity.

2.3. Large language models

Large language models (LLMs) represent a prominent type of artificial intelligence (AI) systems [48], typically built upon Transformer-based neural architectures and trained on extensive general text datasets [49,50]. LLMs typically consist of hundreds of billions or even more parameters, implemented through deep neural networks that stack multiple multi-head self-attention layers [16]. Previous studies have shown that LLMs are capable of delivering remarkable performance among various natural language processing (NLP) tasks without relying on external information [51,52]. Prominent examples, such as GPT-3 and GPT-4, leverage massive textual corpora for their training and have shown their tremendous capabilities to address specialized downstream applications, including language translation and question answering [53].

This also introduces new potential solutions for the NER tasks. One way is to fine-tune the LLM with annotated training data [54]. This requires the LLM to support fine-tuning, and it is preferable for the model to be open-sourced to enable further modifications, as exemplified by open models such as LLaMA [17]. Another way is through their zero-shot and few-shot learning, which enable effective entity recognition even when annotated training data is scarce [19,55]. Through prompt-based extraction, LLMs leverage general linguistic knowledge to infer and identify entities without extensive domain-specific training [55,56]. Shuhe Wang et al. [19] explored the potential of fine-tuning LLMs for NER tasks, as exemplified by GPT-NER, which reframes NER as a text-generation problem. While models such as GPT-3 [16] and its advanced successors (e.g., GPT-4o [57]) have demonstrated impressive few-shot and zero-shot capabilities, their reliance on prompt engineering, rather than traditional fine-tuning, can limit performance on domain-specific tasks. Previous research has focused on this method because many existing LLMs are not open-sourced and do not support self-fine-tuning. For open-source model families, one of the most widely recognized was LLaMA, which Meta AI introduced in 2023 [17]. Since its release, LLaMA has attracted significant attention from academia and the industry. It has demonstrated exceptional performance on numerous open benchmarks and quickly established itself as one of the most popular open-source large language models [49]. Many researchers adopt instruction tuning and fine-tuning techniques to build upon the LLaMA models. Due to its relatively low computational overhead, fine-tuning has emerged as a leading strategy for developing customized or specialized models [49].

Despite these advancements, challenges persist in NER, especially in domain-specific contexts like cybersecurity. The rapid evolution of terminologies, ambiguity of polysemous terms [58], and scarcity of annotated datasets [59] necessitate continuous innovation in methodologies and model architectures. In this context, our proposed CyberLLaMA framework leverages the strengths of LLaMA-3.2-3B, BiLSTM, and CRFs to address these challenges effectively, pushing the boundaries of what NER systems can achieve in specialized domains.

3. Cybersecurity-specific dataset construction

In this study, we develop an updated and comprehensive cybersecurity-specific dataset that contains the most up-to-date cybersecurity articles and terms as our first contribution. Specifically, the dataset comprises 42,404 cybersecurity-related articles, and 4788 unique cybersecurity terms.

Compared to existing cybersecurity NER datasets [25,60,61], our dataset offers several distinct advantages. First, it is built from a more diverse range of sources, including newspapers, security blogs, and official websites, enabling broader linguistic and contextual coverage. Second, it is significantly larger, comprising 42,404 articles. Finally, our dataset includes articles collected up to 2024, ensuring the inclusion of more recent and emerging terminology.

As illustrated in Fig. 1, our dataset building pipeline spans multiple stages, from data collection to final dataset preparation. We begin by aggregating a broad collection of cybersecurity-related articles from three key sources: newspapers, security blogs, and official websites. These articles are initially filtered using predefined cybersecurity keywords and subsequently verified manually to ensure relevance. Next, two complementary approaches are employed to extract candidate cybersecurity terms: one driven by a user study with both IT and non-IT participants, and another using an automatic term identification method that compares term frequencies in cybersecurity versus general corpora. The candidate corpora are then merged, lemmatized, and de-duplicated to form a refined list of 4788 unique security terms. Finally, this security-term corpus guides the annotation of the entire dataset using a BIO scheme (One commonly used method in sequence labeling tasks, particularly for Named Entity Recognition (NER), further details can be found in 3.3), followed by tokenization with a pre-trained LLaMA-based tokenizer and reformatting into a structure compatible with the Hugging Face datasets library. The resulting dataset is split into training and testing sets for subsequent Named Entity Recognition (NER) model development.

In the following subsections, we detail each stage of this pipeline: the Data Source section describes our article collection process, the Data Annotation section explains how security terms are identified and verified, and the Dataset Construction and Preparation section outlines the final processing steps leading to our ready-to-use dataset.

3.1. Data source

The collected articles are in English and sourced from three main outlets: newspapers, security blogs, and official websites. Table 1 presents a representative snippet from each category. Newspaper articles were chosen based on trending topics such as novel cyber attacks, hacking incidents, and online safety measures, with a predefined set of keywords used for initial filtering followed by manual verification to confirm their security focus. Security blogs focus on cybersecurity topics, offering news analysis, technology updates, and insights ranging from succinct summaries to in-depth evaluations designed for readers with diverse levels of expertise. Consequently, these blogs provide the largest volume of articles and the most extensive collection of security-related terminology within our dataset. Meanwhile, official websites are sourced from authorized institutions from industry, academia, and government. These articles provide cybersecurity guidelines and protective measures tailored to their target audiences, such as customers, employees, and

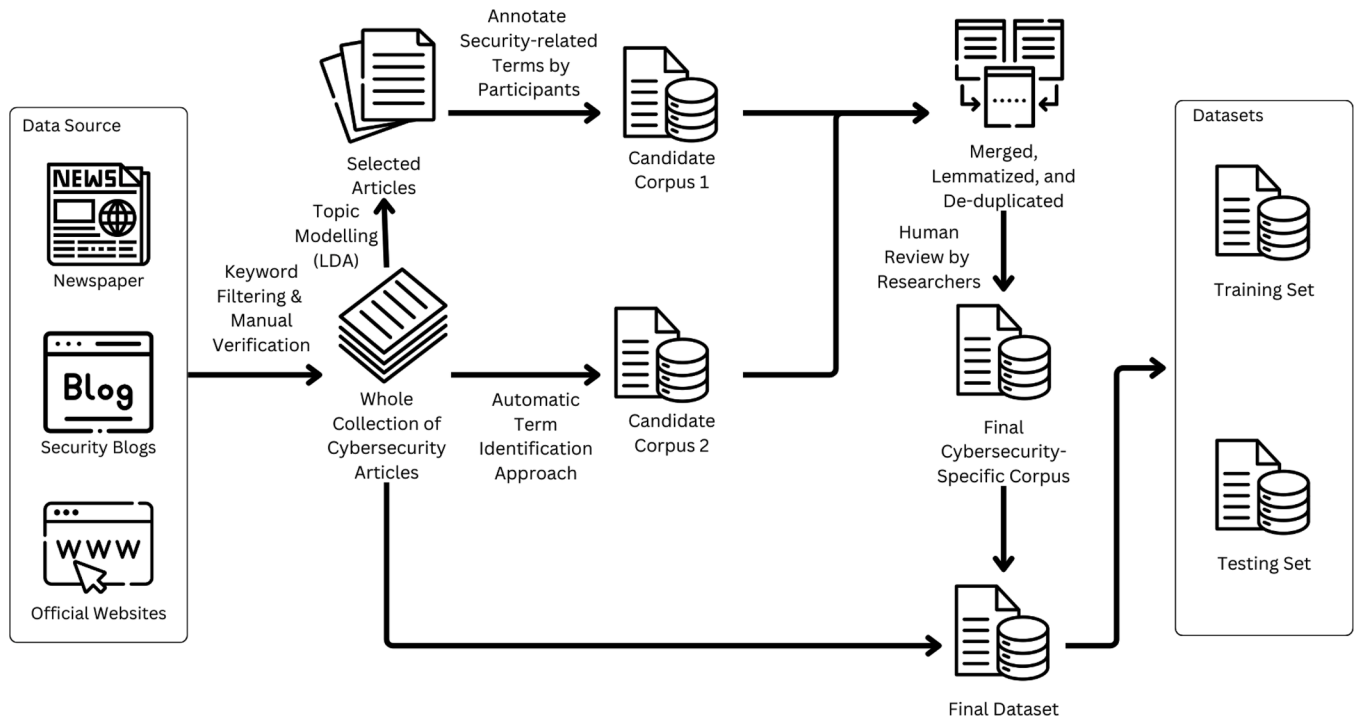


Fig. 1. The dataset building pipeline.

Table 1
Representative excerpts for each data-source category.

Category	Representative excerpt
newspapers	"A Moscow court on Thursday jailed the leader of a notorious hacking group for two years after breaching senior officials' accounts."
security blogs	"In 1968 and 1969, the smart-card was patented in Germany and is now a linchpin of IoT security."
official websites	"Inviting the CFO, COO, and CMO into regular tabletop exercises fosters shared accountability and strengthens enterprise-wide cyber-resilience."

the general public, to maintain their online safety. In addition, many of these sites feature forums where users can discuss relevant security topics, seek advice, and share their experiences. These discussions are also collected as part of our dataset. To ensure that the collected articles were genuinely focused on security, we first applied a set of predefined cybersecurity keywords and then manually reviewed the content to filter out any irrelevant material.

3.2. Data annotation

To support cybersecurity-specific Named Entity Recognition (NER) tasks, we developed a comprehensive security-term corpus via a multi-stage process that annotated all security-related terms in the collected articles. This corpus encompasses both single-word and multi-word phrases, capturing not only frequently occurring terms but also those perceived as ambiguous by users, with case sensitivity preserved throughout.

Initially, based on earlier research [62], a user study was conducted, where participants from both IT and non-IT backgrounds annotated security-related terms in selected texts. Based on the aforementioned research [62], the representative texts are selected through a topic modeling [63] approach using Latent Dirichlet Allocation (LDA) [63]. The LDA was implemented in Python with the gensim library [64], and the LDA model estimated the distribution of topics across documents, with

coherence measures used to ensure the resulting topics were human-interpretable. After experimenting with various numbers of topics, we determined that ten topics achieved the highest coherence. These topics, ranging from government/company reports to network attacks and malicious software, covered the full spectrum of the dataset, from which we randomly selected 20 articles per topic, yielding a final set of 200 representative articles for participants to annotate manually. Through this, an initial candidate corpus was constructed. Meanwhile, we employed an automatic term identification approach [65] to process the dataset of the cybersecurity articles, which quantifies term significance by comparing the relative frequency of words in a domain-specific corpus to that in a general corpus, to build another candidate security-specific corpus. Subsequently, the two candidate security-specific corpora were merged, lemmatized, and de-duplicated. Each term was then traced back to its original context to assess its relevance by two of our researchers, and decisions were made regarding its retention or removal. Cohen's Kappa [66] was employed to evaluate inter-annotator agreement, yielding an impressive score of 0.95 that underscored substantial consensus among the labelers. The resulting refined corpus of 4788 unique security terms (examples see Table 2) subsequently informed our annotation process, where each document was tokenized and labeled using the BIO scheme to ensure consistent, high-quality annotations for model training and evaluation.

Table 2
Examples of the security terms drawn from the 4788-term corpus.

Security-term example
OS-level virtualization
Physical attack
Plug-in Application
Private Network
Self-encrypting deception
Spear phishing
⋮

Table 3
Illustrative example of security term tagging.

Token	Tag
Similar	O
to	O
DNS	B-security
amplification	I-security
attacks	I-security
,	O

3.3. Dataset construction and preparation

Building upon the curated security-term corpus, we then annotated the entire dataset using cutting-edge named entity recognition techniques, drawing on the methodologies from previous research [24,67]. Next, each detected term was transformed into the BIO annotation format, where B-security marks the beginning of a security term or a single-word term, I-security denotes tokens within a security term, and O indicates tokens outside any term, an example could refer to Table 3. This allocation is guided by two considerations. First, our dataset contains

Next, the tagged texts were reformatted into a structure compatible with the Hugging Face datasets library, with each token aligned with its label and sentence boundaries indicated by empty lines. The dataset was then tokenized using a pre-trained LLaMA-based tokenizer (via the AutoTokenizer), ensuring proper sub-word segmentation and label alignment. Finally, we split the data into a training set (90%) and a testing set (10%) for our NER model, CyberLLaMA. This split was chosen for three main reasons. First, due to the large volume of our proposed dataset, the 10% test set still contains around 4240 documents, which is already larger than many existing cybersecurity datasets in total document count reported in prior research [25,68]. Second, a 90/10 split aligns with common practice in large-scale NER benchmarks. Third, following the scaling-law analysis by Hernandez et al. [69], which demonstrates that large language models achieve progressively lower loss, and therefore better downstream performance, as the volume of fine-tuning data increases, we allocate 90% of our corpus to training in order to maximize model performance while retaining a sufficient portion for unbiased evaluation.

4. Methodology of CyberLLaMA

In this section, we provide an in-depth discussion of CyberLLaMA's architecture, data annotation process, tokenization and label alignment, sequence modeling using a Bidirectional Long Short-Term Memory (BiLSTM) network and Conditional Random Fields (CRF), and the fine-tuning strategy employed to enhance its performance.

4.1. Model overview

We present CyberLLaMA, a cybersecurity-specific Named Entity Recognition (NER) framework that integrates a fine-tuned Large Language Model (LLM), LLaMA-3.2-3B with a Conditional Random Field (CRF) layer. Based on the pre-trained LLaMA model, CyberLLaMA is designed to address challenges in cybersecurity NER tasks, such as handling domain-specific jargon, abbreviations, and evolving terminology. Fig. 2 illustrates the system's architecture, where a BiLSTM layer augments the fine-tuned LLaMA representations and a CRF layer enforces structured sequence predictions, together delivering robust NER performance. Algorithm 1 details the unified training and inference procedure that underpins the model.

CyberLLaMA utilizes the contextual understanding capabilities of LLaMA, fine-tuned with a cybersecurity-specific dataset to specialize in identifying and classifying security-related entities. To enhance sequence labeling performance, a CRF layer is incorporated, allowing the

model to capture dependencies between labels and produce contextually consistent predictions. This combination ensures that the model effectively recognizes complex entity structures common in cybersecurity texts, such as "Advanced Persistent Threat" (APT) or "Distributed Denial of Service" (DDoS).

Algorithm 1: CyberLLaMA - unified training and inference.

Input: Labeled dataset D , number of epochs E , batch size B , optional input sentence x

Output: Fine-tuned parameters θ^* , optional label sequence y for x

```

1 Initialisation:
2  $\theta \leftarrow \theta_0$ ; // pre-trained LLaMA encoder + randomly initialised task heads
3 Training phase:
4 for  $e \leftarrow 1$  to  $E$  do
5   Shuffle  $D$ ;
6   foreach mini-batch  $B \subset D$  of size  $B$  do
7      $w \leftarrow \text{Tokenizer}(B)$ ; // tokenise inputs
8      $z \leftarrow \text{LLaMA}_\theta(w)$ ; // contextual embeddings
9      $h^{(0)} \leftarrow \text{Dropout}(z)$ ;
10     $h^{(1)} \leftarrow W_{\text{proj}} h^{(0)} + b_{\text{proj}}$ ; // projection
11     $h \leftarrow \text{BiLSTM}(h^{(1)})$ ;
12     $s \leftarrow W_{\text{cls}} h + b_{\text{cls}}$ ; // emission scores
13     $\mathcal{L} \leftarrow -\log P_{\text{CRF}}(y | s)$ ; // CRF negative log-likelihood
14     $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$ ; // AdamW step, lr  $\eta$ 
15 Inference phase:
16 if sentence  $x$  is provided then
17    $w \leftarrow \text{Tokenizer}(x)$ ;
18    $z \leftarrow \text{LLaMA}_{\theta^*}(w)$ ;
19    $h^{(0)} \leftarrow \text{Dropout}(z)$ ; // inactive at eval
20    $h^{(1)} \leftarrow W_{\text{proj}} h^{(0)} + b_{\text{proj}}$ ;
21    $h \leftarrow \text{BiLSTM}(h^{(1)})$ ;
22    $s \leftarrow W_{\text{cls}} h + b_{\text{cls}}$ ;
23    $y \leftarrow \text{Viterbi}(s)$ ; // argmax decoding
24 return  $\theta^*, y$ 

```

4.2. Problem formalisation

In our work, we formalise the cybersecurity named entity recognition (NER) task as follows. Let D denote a collection of cybersecurity documents, where each document $D \in D$ is represented as a sequence of tokens:

$$D = \{w_1, w_2, \dots, w_n\}.$$

Each token w_i is assigned a label $y_i \in \{O, B\text{-SECURITY}, I\text{-SECURITY}\}$ according to the BIO tagging scheme [70], where 'O' indicates that the token is outside a security-related term, 'B-SECURITY' indicates the beginning of a security term (or that the term consists of a single token), and 'I-SECURITY' indicates that the token is inside a security term. Our objective is to learn a mapping function $f(\cdot)$, parameterised by a fine-tuned LLaMA-based model that incorporates a Bi-LSTM for sequential modeling and an integrated Conditional Random Field (CRF) layer for structured prediction, that predicts the most likely label sequence y for any input token sequence:

$$\hat{y} = f(\{w_i\}) = \arg \max_y P(y | \{w_i\}).$$

Here, the probability $P(y | \{w_i\})$ is modelled by combining deep contextual embeddings from a pre-trained language model, the sequential modeling capabilities of a Bi-LSTM, and the structured prediction benefits of the CRF layer.

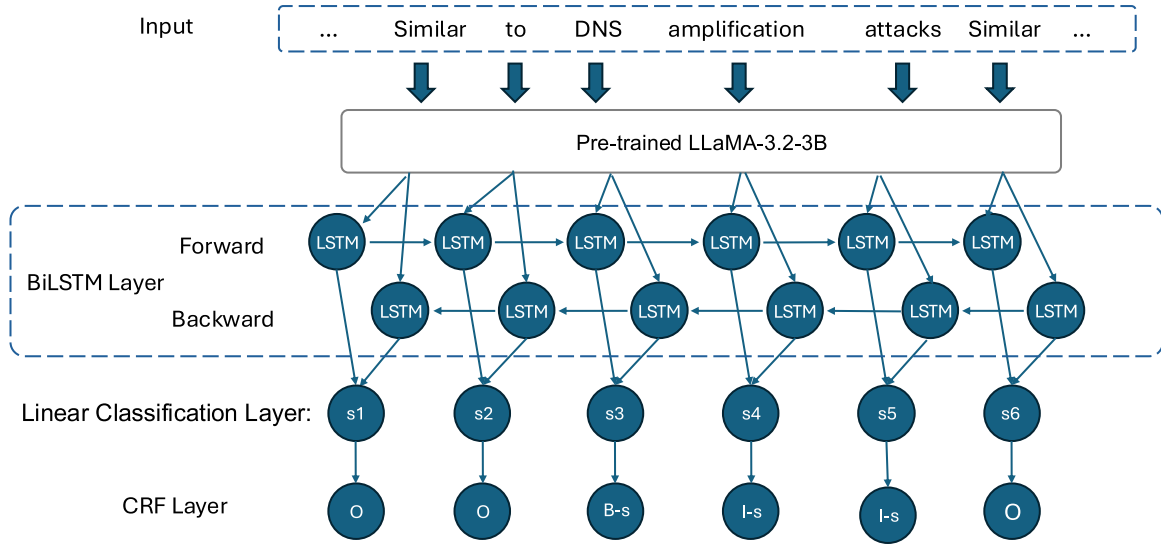


Fig. 2. The architecture of CyberLLaMA.

4.3. Tokenization and label alignment

After annotation, the documents are processed using sub-word tokenisation [43] to generate inputs compatible with our transformer model. We employ the AutoTokenizer [71] to split each annotated document into sub-word tokens while preserving the original word boundaries. A critical challenge in this step is aligning the original BIO labels with the sub-word tokens. This is achieved by generating a word-to-token mapping for each example and assigning the correct label, or a placeholder value of -100 for tokens that do not correspond to any original word, such as padding tokens, to each sub-token. This label alignment ensures that the model receives consistent and accurate signals during training.

4.4. Integrating LLaMA, BiLSTM, and CRF

CyberLLaMA leverages a hybrid architecture that combines the rich, deep contextual embeddings of a fine-tuned LLaMA model with the sequential modeling power of a Bidirectional Long Short-Term Memory (BiLSTM) network and the structured prediction capabilities of a Conditional Random Field (CRF).

Detailed Component Principles. We first provide an in-depth explanation of CyberLLaMA's core components (BiLSTM and CRF) to better explain how these layers contribute to the overall performance of the model.

Bidirectional LSTM (BiLSTM): This component is designed to capture sequential dependencies from both directions in the token sequence, which is crucial for Named Entity Recognition (NER) tasks where the context surrounding a token influences its label. A standard LSTM cell computes its hidden state using the following equations:

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i),$$

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f),$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o),$$

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c),$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t,$$

$$h_t = o_t \odot \tanh(c_t),$$

where σ represents the sigmoid function and $\odot$ denotes element-wise multiplication. The final representation for each token is the concatenation of the forward and backward hidden states.

Conditional Random Field (CRF): The CRF layer models the entire output label sequence jointly rather than predicting each token's label independently. It computes a score for a label sequence using

$$\psi(y, s) = \sum_{i=1}^n [\phi(y_i, s_i) + \tau(y_{i-1}, y_i)].$$

The probability of a particular label sequence is then given by

$$P(y | w) = \frac{\exp(\psi(y, s))}{\sum_{y'} \exp(\psi(y', s))},$$

and the model is trained by minimizing the negative log-likelihood loss:

$$\mathcal{L} = -\log P(y | w).$$

During inference, the Viterbi algorithm is used to determine the label sequence $\hat{y}$ that maximizes the score $\psi(y, s)$.

Breakdown of Contributions. In the following we provide a detailed explanation of how each component contributes to the overall performance.

Contextual Embedding: The pre-trained LLaMA model processes the tokenized input w to generate contextual embeddings (or initial logits) for each token:

$$z = \text{LLaMA}(w).$$

These embeddings capture rich semantic and syntactic information from the input text.

Dropout and Linear Projection: To mitigate overfitting and improve generalization, a dropout layer is applied to z . Subsequently, a linear projection transforms the output dimension from the number of labels to the hidden size expected by the BiLSTM:

$$h^{(0)} = \text{Dropout}(z), \quad h^{(1)} = W_{\text{proj}} h^{(0)} + b_{\text{proj}}.$$

This step aligns the features into a space where sequential patterns can be more effectively modeled.

BiLSTM for Sequential Modeling: The projected features $h^{(1)}$ are fed into a Bidirectional LSTM. The forward LSTM computes

$$\bar{h}_i = \text{LSTM}_{\text{forward}}(h_i^{(1)}, \bar{h}_{i-1}),$$

while the backward LSTM computes

$$\bar{h}_i = \text{LSTM}_{\text{backward}}(h_i^{(1)}, \bar{h}_{i+1}).$$

The final hidden state for each token is obtained by concatenating these two outputs:

$$h_i = [\bar{h}_i; \bar{h}_i].$$

This concatenation enables the network to capture long-range dependencies and context from both directions.

Linear Classification Layer: The output of the BiLSTM is then passed through a linear layer that maps the hidden state back to the label space:

$$s_i = \mathbf{W}_{\text{cls}} \mathbf{h}_i + \mathbf{b}_{\text{cls}},$$

where s_i represents the refined logits (emission scores) for token i .

CRF Integration: The CRF layer models the joint probability of the output label sequence by considering both the emission and transition scores. The overall sequence score is given by

$$\psi(\mathbf{y}, \mathbf{s}) = \sum_{i=1}^n [\phi(y_i, s_i) + \tau(y_{i-1}, y_i)],$$

where $\phi(y_i, s_i)$ is the emission score derived from the classifier's output for token i , and $\tau(y_{i-1}, y_i)$ is a learned parameter reflecting the likelihood of transitioning from label y_{i-1} to label y_i . The conditional probability of a label sequence is defined as

$$P(\mathbf{y} | \mathbf{w}) = \frac{\exp(\psi(\mathbf{y}, \mathbf{s}))}{\sum_{\mathbf{y}'} \exp(\psi(\mathbf{y}', \mathbf{s}))},$$

and the model is trained by minimizing the negative log-likelihood loss:

$$\mathcal{L} = -\log P(\mathbf{y} | \mathbf{w}) = -\log \frac{\exp(\psi(\mathbf{y}, \mathbf{s}))}{\sum_{\mathbf{y}'} \exp(\psi(\mathbf{y}', \mathbf{s}))}.$$

During inference, the Viterbi algorithm is used to decode the most probable label sequence:

$$\hat{\mathbf{y}} = \arg \max_{\mathbf{y}} \psi(\mathbf{y}, \mathbf{s}).$$

4.5. Feature projection and regularisation

Given contextual embeddings $\mathbf{z} \in \mathbb{R}^{n \times d_{\text{model}}}$ from the LLaMA encoder, we first apply dropout with rate p (tuned on the validation set). A linear map $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{d_{\text{proj}} \times d_{\text{model}}}$ then compresses these representations into a d_{proj} -dimensional sub-space:

$$\mathbf{h}^{(1)} = \mathbf{W}_{\text{proj}} \text{Dropout}(\mathbf{z}) + \mathbf{b}_{\text{proj}}.$$

The reduced vectors are subsequently fed to a bidirectional LSTM with d_{hid} hidden units per direction, trimming recurrent parameters while focusing the sequence-labelling stack on features most salient to the BIO objective. This transformation effectively recasts the traditional sequence-labeling problem as a conditional generation task, in which the model generates a coherent BIO tag sequence conditioned on the input tokens.

Overall, CyberLLaMA benefits from this hybrid architecture by combining deep contextual embeddings from the LLaMA model, the sequential modeling power of the BiLSTM, and the structured prediction capabilities of the CRF layer. This approach is particularly effective for cybersecurity-related Named Entity Recognition tasks, where both detailed context and structured output are essential.

5. Evaluation

In this section, we describe the experimental setup, which includes the details of the dataset used for training and evaluation, as well as the environment in which the experiments were conducted. Additionally, we present the results of our evaluation, highlighting the performance of CyberLLaMA on cybersecurity-specific Named Entity Recognition (NER) tasks.

5.1. Experiment setup

Implementation details of CyberLLaMA. Fine-tuning CyberLLaMA involves adapting the pre-trained LLaMA model to cybersecurity-specific Named Entity Recognition (NER) tasks by integrating additional architectural components, optimizing hyperparameters, and leveraging

high-performance hardware resources to maximize model efficiency and performance. 1) **Model Initialization:** The pre-trained LLaMA model is initialized with its original weights, and a Conditional Random Field (CRF) layer is added to improve sequence labeling by modeling the dependencies between output labels, ensuring consistent and contextually accurate predictions. 2) **Architecture Enhancements:** To refine the model's contextual understanding, a bi-directional LSTM layer is incorporated following the base model's output to capture sequential dependencies and enhance the representation of cybersecurity-specific text, while a dropout layer with a probability of 0.2 is applied to mitigate overfitting during training. 3) **Domain-Specific Tokenization:** Word-Piece tokenization is employed to preprocess the dataset, addressing gaps in the LLaMA vocabulary and ensuring comprehensive coverage of domain-specific terms. 4) **Training Configuration:** The fine-tuning process is conducted using two NVIDIA A100 GPUs, each with 80GB of memory, which enables efficient parallelization; the model is trained for 5 epochs with a batch size of 4, a learning rate of $2e-5$, and a cosine learning rate scheduler, with mixed-precision training enabled to balance memory usage and computational efficiency. 5) **Monitoring and Optimization:** GPU utilization is continuously monitored using NVIDIA's `nvidia-smi` tool, logging metrics such as utilization, memory usage, and free memory every 30 min; these logs indicate that GPU usage during fine-tuning remained consistently high, reaching 100% utilization for both GPUs with memory usage peaking at approximately 81GB.

Metrics. Performance evaluation is crucial for the development and improvement of NER systems, as it provides essential insights that guide model enhancements and enable effective benchmarking against baseline models. To comprehensively evaluate CyberLLaMA's performance on cybersecurity-specific NER tasks, we utilize widely adopted metrics, including F1 score, precision, and recall, which enable us to benchmark our model against selected baseline models.

Experiment Environment. Experiments were conducted on a Rocky Linux 9.5 (Blue Onyx) server equipped with two NVIDIA A100-SXM4 GPUs, each providing 80GB of VRAM. The software environment included Python 3.9.21, PyTorch 2.4.1 + cu121, CUDA 12.7, cuDNN 9.0.1, and Hugging Face Transformers 4.45.2. This configuration ensured compatibility and efficiency for the fine-tuning and evaluation of the CyberLLaMA model. Detailed configurations are discussed in Section 5.6.

For readers' convenience and to facilitate exact replication, we now provide all hardware, software, and hyper-parameter settings in Table 4.

Baselines. In our experiments, we compare our proposed CyberLLaMA framework with a diverse set of state-of-the-art baselines spanning four categories: general-purpose pretrained models, domain-specific pretrained models, neural architectures augmented with lexical representations, and prompt-based large language models.

General-purpose pretrained models. These models, such as BERT-Base [43] and RoBERTa [36], are pretrained on large-scale general corpora. BERT leverages bidirectional contextual embeddings through masked language modeling, while RoBERTa enhances BERT's training

Table 4
Reproducibility settings for CYBERLLaMA fine-tuning.

Category	Setting / value
Hardware	2 × NVIDIA A100-SXM4 (80 GB each); Rocky Linux 9.5; mixed-precision enabled
Software	Python 3.9.21; PyTorch 2.4.1 + cu121; CUDA 12.7; cuDNN 9.0.1; Hugging Face Transformers 4.45.2
Per-device batch	4
Gradient accumulation	4 ⇒ effective batch = 16
Max training steps	5960
Epochs	5
Learning rate	2×10^{-5} ; cosine scheduler
Weight decay	0.01
Logging frequency	every 100 steps
Evaluation cadence	evaluation / checkpoint every 500 steps

dynamics with dynamic masking and larger batch sizes. Both models were fine-tuned on our cybersecurity dataset to assess their ability to adapt to domain-specific jargon despite lacking specialized pretraining and structured prediction mechanisms (e.g., CRF) for resolving ambiguous entity types and enforcing label consistency.

Domain-specific pretrained models. This category includes models specifically adapted to the cybersecurity domain. Notably, SCBERT [72] is a variant of BERT pretrained on security-related texts and fine-tuned with a CRF layer, representing the prior state-of-the-art in cybersecurity NER. Additionally, models like BioBERT [38] and SciBERT [37], pretrained on biomedical or scientific corpora respectively, are included to evaluate whether domain-specific pretraining from related fields can capture cybersecurity terminology.

Neural architectures with lexical feature augmentation. This category includes a model that enhances neural sequence labeling by incorporating automatically generated lexical features [73]. In particular, the BiLSTM-CRF model with Lexical Similarity (LS) Features augments the standard BiLSTM-CRF by integrating 120-dimensional LS vectors computed offline via distant supervision.

Prompt-based large language models. Recognizing the recent trend in leveraging powerful LLMs for NER in a zero-shot or few-shot manner, we include GPT-NER [19] as a baseline. This prompt-based approach specially designed prompts that incorporate few-shot demonstrations to guide the model in identifying and labeling entities. In our experiments, we evaluate GPT-NER on our proposed dataset for identifying cybersecurity-specific terms using both GPT-4 and GPT-4o, thereby assessing its effectiveness in identifying cybersecurity-specific terms

The inclusion of these baselines enables a comprehensive evaluation of CyberLLaMA. Comparisons with models such as BERT-Base + CRF + Fine-Tune and SCBERT elucidate the benefits of integrating a fine-tuned LLaMA backbone with CRF-based sequence modeling, while the analysis of the LS-augmented BiLSTM-CRF model highlights the complementary strengths of dynamic contextual embeddings and static lexical representations. Evaluating GPT-NER in a prompt-based setting further underscores the advantages of domain-specific fine-tuning and architectural innovations in addressing the unique challenges of cybersecurity NER.

5.2. Specific comparison on shared dataset

In Table 5, we compare the performance of several state-of-the-art NER models on our cybersecurity dataset. Among the BERT-based approaches, SCBERT with fine-tuning achieves an F1 score of 85.47 %, substantially outperforming both Bert-Base + CRF + Fine-Tune (66.80 %) and Bert-Base + Fine-Tune (66.35 %). Domain-specific models such as BioBERT and SciBERT yield F1 scores of 65.67 % and 63.19 %, respectively, indicating that while pretraining on biomedical or scientific text offers some advantage, it is less effective than models specifically adapted for security text. In contrast, RoBERTa [36] + Fine-Tune per-

forms even worse (56.59 % F1), suggesting that general-purpose language models struggle with cybersecurity jargon. Similarly, models leveraging additional lexical features, like the BiLSTM-CRF + Lexical Features model (51.70 % F1), fail to capture the complexity of the domain. Notably, our proposed CyberLLaMA framework achieves an F1 score of 98.88 %, with nearly perfect precision (98.62 %) and recall (99.13 %). This represents an absolute improvement of over 13 percentage points in F1 compared to SCBERT and highlights the effectiveness of integrating a fine-tuned LLaMA model with CRF-based sequence modeling tailored for cybersecurity NER.

5.3. Baseline comparison with GPT-NER

To further substantiate the necessity for domain-specific adaptation and fine-tuning of LLMs for NER tasks, we evaluated a general-purpose prompt-based baseline-GPT-NER [19], along with a GPT-4o zero-shot approach and our proposed model, CyberLLaMA, on a cybersecurity-specific dataset. We followed the GPT-NER methodology by employing its three-part prompt design-which comprises a task description, few-shot demonstrations, and an input sentence-supplemented by the paper's entity-level embedding retrieval strategy to ensure that the selected examples closely align with the input context. Using GPT-4o as the base model, GPT-NER achieved a Precision of 54.9 %, Recall of 50.7 %, and an F1 Score of 52.7 %, significantly outperforming the zero-shot GPT-4o approach (F1 Score: 30.0 %, Precision: 25.0 %, Recall: 37.5 %). In contrast, our fine-tuned model, CyberLLaMA, attained an F1 Score of 98.88 %, a Precision of 98.62 %, and a Recall of 99.13 %.

These results indicate that while few-shot approaches, enhanced by effective example retrieval strategies, can substantially improve performance over zero-shot methods, they still fall short in domain-specific NER tasks, particularly in the cybersecurity domain. Our approach, which leverages fine-tuning to capture the nuanced terminology and context inherent in cybersecurity data, ultimately yields far superior performance.

5.4. Ablation study

To further analyze the contributions of individual components within CyberLLaMA, we conducted an ablation experiment by progressively removing the CRF layer and the BiLSTM layer from the model architecture. The results of this experiment are summarized in Table 6.

The results in Table 6 demonstrate the significant impact of the CRF and BiLSTM layers on the model's performance across all evaluation metrics. **First**, when we keep the BiLSTM layer but remove the CRF layer, the F1 score drops from 98.88 % to 95.48 % (a decrease of 3.40 percentage points), with precision and recall falling to 95.20 % and 95.76 %, respectively. **Second**, removing *both* the CRF and BiLSTM layers leads to a further decline in performance: the F1 score falls to 93.61 %, precision to 93.01 %, and recall to 94.23 %. Overall, the inclusion of the CRF layer alone contributes roughly 3.4 points of the total 5.27-point F1 improvement, while the BiLSTM layer accounts for the remaining gain, highlighting that the two components provide complementary benefits.

In conclusion, the ablation study confirms that the CRF layer and the BiLSTM layer together significantly enhance CyberLLaMA's performance in cybersecurity Named Entity Recognition (NER). The findings

Table 5
Comparison of NER performance across models on the shared dataset.

Model	F1 Score	Precision (%)	Recall (%)	Accuracy (%)
Bert-Base [43] + CRF [30] + Fine-Tune	66.80	67.21	66.32	96.49
Bert-Base [43] + Fine-Tune	66.35	66.25	66.45	96.42
BioBERT [38] + Fine-Tune	65.67	66.43	64.96	96.39
SciBERT [37] + Fine-Tune	63.19	63.21	63.17	96.09
RoBERTa [36] + Fine-Tune	56.59	62.33	51.81	95.77
BiLSTM-CRF + Lexical Features [73]	51.70	57.24	47.22	95.32
SCBERT [72] + Fine-Tune	85.47	85.79	85.16	98.46
GPT-4o [57]	30.00	25.00	37.50	90.70
GPT-Ner (GPT-4o) [19]	52.70	54.90	50.70	95.16
CyberLLaMA (Proposed)	98.88	98.62	99.13	99.88

Table 6
Comparison of NER performance with and without the CRF layer.

Model	F1 Score	Precision (%)	Recall (%)	Accuracy (%)
CyberLLaMA (w/ CRF&BiLSTM)	98.88	98.62	99.13	99.88
CyberLLaMA (w/ BiLSTM, w/o CRF)	95.48	95.20	95.76	99.52
CyberLLaMA (w/o CRF&BiLSTM)	93.61	93.01	94.23	99.32

validate our design choice and emphasize the importance of structured sequence-modeling techniques for cybersecurity-specific NER tasks.

As a sum up, **CyberLLaMA** achieved an F1 score of 98.88 %, a precision of 98.62 %, and a recall of 99.13 %, underscoring the effectiveness of leveraging fine-tuned large language models with CRF and BiLSTM layers for enhancing cybersecurity NER.

5.5. Experiment with additional dataset

Following the data-collection procedure described in Section 3.1, we assembled an additional test set of **200** recent articles, evenly split among the aforementioned sources. On this set, **CYBERLLAMA** achieves an F_1 of **97.82 %**, **97.12 %** precision, and **98.53 %** recall, further confirming the model's robustness and effectiveness.

5.6. Hyperparameter sensitivity and model optimization

To assess the robustness of *CyberLLaMA* under different training configurations and evaluate the impact of various hyperparameter settings, we conducted two controlled experiments. In our experiments, the differences between the two configurations are reflected in the hyperparameter settings summarized in Table 7. We refer to these configurations as **CyberLLaMA (Baseline)** and **CyberLLaMA (Optimized)**. While the original changes were primarily in training configuration (such as batch size, maximum training steps, weight decay, and LR scheduler), the Optimized configuration also incorporates refinements in our handling of token label alignment and domain-specific terminology standardization, which we hypothesize contribute to the slight performance gains.

Table 7 presents the key training parameters for both configurations, while Table 8 compares their performance metrics. In the Baseline, the effective batch size is 12 (per-device train batch size of 6 with gradient accumulation steps of 2) and the training proceeds for 7950 steps. In contrast, the Optimized configuration uses a per-device train batch size of 4 with gradient accumulation steps of 4, yielding an effective batch size of 16, and the total training steps are reduced to 5,960. In addition, the Optimized configuration applies a cosine LR scheduler with a lower weight decay (0.01 vs. 0.03) and a more frequent logging (every 100 steps) and evaluation (every 500 steps) strategy. These changes are intended to balance the computational overhead from the refined processing techniques with improved convergence properties.

A detailed comparison of these configurations reveals several key observations:

Table 7

Key Training configuration comparison: Baseline vs. Optimized.

Parameter	Baseline	Optimized
Effective Batch Size	12	16
Max Training Steps	7950	5960
Learning Rate	2×10^{-5}	2×10^{-5}
Weight Decay	0.03	0.01
LR Scheduler	linear	cosine
Number of Training Epochs	5	5

Table 8

Performance comparison: CyberLLaMA (Baseline) vs. CyberLLaMA (Optimized).

Metric	Baseline	Optimized	Improvement
F1 Score	98.75 %	98.88 %	+0.13 %
Precision	98.80 %	98.62 %	-0.18 %
Recall	98.69 %	99.13 %	+0.44 %
Accuracy	99.87 %	99.88 %	+0.01 %
Final Training Loss	0.8636	0.8058	-6.7 %
Throughput (Samples/sec)	8.48	6.57	-22.5 %

- **Overall Performance:** The Optimized configuration achieves a marginally higher F1 score (+0.13 %) than the previous setting, primarily due to an increase in recall (+0.44 %) despite a slight drop in precision (-0.18 %). Both configurations reach an nearly identical accuracy, indicating that the improvements mainly affect the identification of relatively rare entity tokens.
- **Configuration Modifications:** The Optimized configuration uses a smaller per-device batch size with higher gradient accumulation (yielding an effective batch size of 16) and fewer training steps (5,960 vs. 7,950), along with a cosine learning rate scheduler and lower weight decay. These modifications likely contribute to more stable training (as indicated by a lower final training loss) and improved recall, suggesting a better balance between regularization and convergence.
- **Throughput and Efficiency:** The Optimized configuration processes fewer samples per second due to its additional processing overhead from refined token alignment and terminology standardization, as well as the change in evaluation/logging frequency. This trade-off between computational efficiency and model performance is evident from the throughput differences.

In summary, the Optimized configuration of *CyberLLaMA* demonstrates slightly improved NER performance (higher recall and marginally higher F1 score) and a lower training loss compared to the Baseline, at the cost of reduced throughput. Given the extremely similar overall performance, both configurations remain viable for deployment, with the choice depending on specific resource and efficiency requirements.

6. Discussion

While **CyberLLaMA** demonstrates state-of-the-art performance in cybersecurity-specific Named Entity Recognition (NER), we critically analyze its limitations and practical implications below.

6.1. Limitations

Scalability and Security Considerations. Our framework leverages LLaMA-3.2-3B, a smaller variant of the LLaMA series. While this choice balances performance and accessibility, larger models like LLaMA-3.2-11B or LLaMA-3.3-70B—though computationally prohibitive for our setup—might better handle rare or contextually ambiguous terms (e.g., polymorphic malware names). Future work should explore quantization or distillation techniques to bridge this gap without sacrificing efficiency. Additionally, recent research indicates that despite the impressive performance of LLMs, their security implications in sensitive applications remain a critical concern [74,75], underlining the need for further investigation in this area.

Dataset Temporal Bias. Our training corpus focuses on post-2000 cybersecurity terminology, omitting historical terms. While LLMs like LLaMA partially mitigate this via pretraining on diverse texts, explicit inclusion of legacy terms could enhance robustness for retrospective threat analysis.

Potential Coverage Gap. Although our dataset covers a diverse range of sources, some cybersecurity subdomains or sources may not be adequately represented, leaving the model's performance in these areas uncertain. Evaluating CyberLLaMA on additional sources representing newly emerging or specialized areas could provide further insights into its robustness and adaptability.

Architectural Constraints. Our proposed framework, which integrates a Bi-LSTM layer, a CRF layer, and a large language model (specifically, LLaMA-3.2-3B), has effectively addressed domain-specific NER tasks, particularly in cybersecurity. However, as recent literature [76] indicates, the integration of large language models into cybersecurity frameworks is still in its early stages, revealing significant potential for more tailored approaches. Other architecture like combining large language models with emerging techniques, such as graph neural networks

or dynamic span attention, may also enhance performance in complex cybersecurity contexts.

6.2. Practical applications & ethical considerations

Practical applications. (1) CyberLLaMA's lightweight makes it easy for client-side deployment. A browser, PDF reader, or social media extension can invoke the model locally to highlight security terms directly on the page, display concise definitions, and link to authoritative references, lowering the jargon barrier for end users and fostering better day-to-day security practices. (2) At platform scale, the same model can run server-side to continuously scan user-generated content for previously unseen threat indicators (e.g., novel malware names or exploit acronyms), flagging for rapid fact-checking or mitigation.

Ethical considerations. (1) All our training articles were exclusively collected from publicly sources and no personal identifiers were included, ensuring that the released checkpoints contain no memorised personally identifiable information. (2) CYBERLLAMA may yield false positives that fuel alert fatigue and false negatives that overlook critical indicators, we position the model as an assistive reference rather than a sole decision-maker, and urge practitioners to combine its outputs with expert judgment for all high-stakes security actions.

6.3. Future directions

While CyberLLaMA shows promise for cybersecurity NER, there remain multiple avenues for further exploration:

- **Scaling the Model:** Larger models, such as *LLaMA-3.2-11B* or *LLaMA-3.3-70B*, as well as open-source alternatives like *deepseek-v3*, may capture rarer or more ambiguous cybersecurity entities. However, deploying such models is often constrained by hardware limitations. Future work could investigate quantization or model distillation to balance computational efficiency with improved coverage of domain-specific terminologies.
- **Other Domain-Specific Adaptations:** To further evaluate CyberLLaMA's adaptability, fine-tuning on other specialized domains (e.g., business & finance, biomedical) could provide deeper insights into its performance under diverse conditions. However, datasets in corresponding fields may need to be constructed.
- **Alternative Architectures:** While CRF layers effectively capture label dependencies, future research might explore novel techniques such as graph neural networks for entity-relation extraction or dynamic span attention for nested entities. These methods could offer more nuanced modeling of complex cybersecurity texts, especially where hierarchical or overlapping entity structures arise.

7. Conclusion

In this paper, we make two primary contributions. Our first contribution is an updated and enriched cybersecurity dataset that integrates over 42,404 articles from multiple sound sources. This dataset incorporates recent and diverse terminologies to accurately reflect the evolving threat landscape, providing a robust foundation for domain-specific research and NER tasks in cybersecurity.

Our second contribution is CyberLLaMA, a state-of-the-art cybersecurity-specific Named Entity Recognition framework. CyberLLaMA is built by fine-tuning the LLaMA-3.2-3B model and enhancing it with a Conditional Random Field (CRF) layer and a BiLSTM layer to effectively address the challenges posed by domain-specific jargon, abbreviations, and complex contextual dependencies in cybersecurity texts. Experimental evaluations show that CyberLLaMA achieves state-of-the-art performance, significantly outperforming existing models such as SCBERT and RoBERTa with an F1 score of 98.88 %.

Despite hardware limitations restricting our exploration to smaller LLMs, we believe CyberLLaMA effectively represents the capabilities

of large language models while maintaining accessibility. The updated dataset also strengthens domain relevance, although historical term coverage remains an area for improvement. Future work could explore larger models, more extensive corpora, and additional subdomains to further enhance performance and applicability.

Overall, CyberLLaMA sets a new benchmark for cybersecurity-specific NER, offering a practical and scalable solution for cybersecurity professionals and researchers. Its success demonstrates the potential of leveraging fine-tuned LLMs for domain-specific tasks, paving the way for further advancements in cybersecurity and beyond.

CRedit authorship contribution statement

Hao Zhang: Writing – original draft; **Tingmin Wu:** Validation; **Tianqing Zhu:** Writing – review & editing, Conceptualization; **Sheng Wen:** Conceptualization; **Yang Xiang:** Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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